

SOLAR METRO TRAINSET

DRIVERLESS. SIMPLE. SMART.

- 3 CAR TRAINSET
- DRIVERLESS – FULLY AUTOMATED
- NO CATENARY – BATTERY POWERED
- SOLAR PANELS ON ROOF
- OPEN GLASS DRIVERLESS ENDS
- LED HEADLAMPS & DRL MARKER BARS
- TWO DOUBLE-LEAF DOOR PAIRS PER SIDE PER CAR
- STANDARD BOGIES (~3 m LONG)
- ~3 m HIGH-FLOOR BOGIE-END DECKS
- ~10 m LOW-FLOOR CENTRE SECTION
- WIDE PASSENGER WINDOWS (COTS)
- SODIUM-ION BATTERIES UNDER SEATS
- DESIGNED FOR 1 km STATION SPACING



TRAINSET CONFIGURATION (3 CARS)

CAR 1 – POWERED (DRIVERLESS END)



CAR 2 – TRAILER (LOW FLOOR)

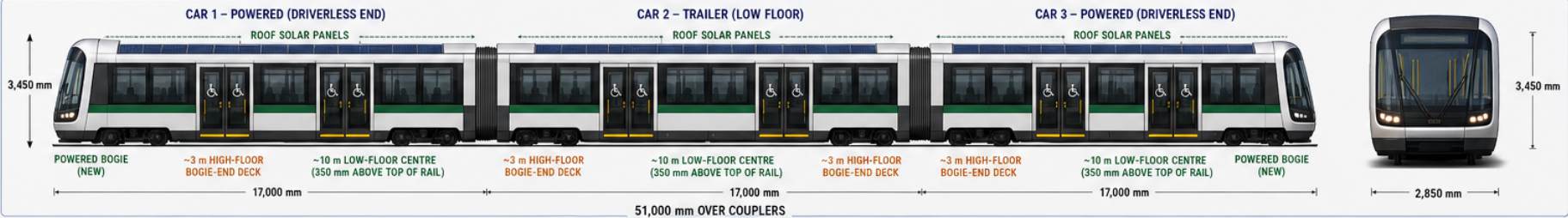


CAR 3 – POWERED (DRIVERLESS END)

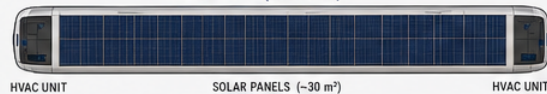


P = POWERED BOGIE (NEW)

T = TRAILER BOGIE (CONVERTED FREIGHT)



ROOF PLAN (EACH CAR)



BOGIE STRATEGY



NEW POWERED BOGIE



- Reused Bogie Frame
- New Bearings
- New Disc Brakes
- Air Springs (Secondary)
- Dampers & Anti-Roll Bar
- Speed Sensor

- 2 x 150 kW Motors
- Gearboxes
- Disc Springs
- Air Springs (Primary & Secondary)
- Traction Inverters Above

INTERIOR VIEW (TYPICAL)



2 DOOR PAIRS PER SIDE PER CAR



INTERIOR – SEAT OVER BATTERY (TYPICAL)



TYPICAL CROSS SECTION



KEY SPECIFICATIONS (PER 3-CAR TRAINSET)

- Total Length (over couplers): 51,000 mm
- Car Length: 17,000 mm
- Car Width: 2,850 mm
- Car Height: 3,450 mm
- Floor Heights: 350 mm (low-floor centre) 760 mm (high-floor ends)
- Total Batteries: ~450 kWh (150 kWh per car)
- Traction Power: 600 kW (4 motors total)
- Max Speed: Max Speed: 90 km/h
- Passenger Capacity: ~450 (150 per car)
- Designed for 1 km Station Spacing

ENERGY SYSTEM



WHY THIS DESIGN IS LOW COST

- ✓ No driver cabs = more space, lower cost
- ✓ No catenary = lower infrastructure cost
- ✓ Simple box design = easy to build & maintain
- ✓ Reuse freight bogies = major savings
- ✓ Batteries under seats = efficient & safe
- ✓ Solar on roof = reduced energy cost

IDEAL FOR

- Urban / Peri-urban Corridors
- Developing Cities
- Low Traffic Metro Lines
- Rapid Deployment

1. The Problem

Urban rail is too slow and expensive to deploy

- Proprietary vehicle and signalling packages create lock-in.
- Capital cost makes medium-size corridors hard to justify.
- Integration evidence is often document-heavy and hard to regenerate.
- Local manufacturers are pushed toward low-value installation work.

OpenSourceRail's bet

- Publish reference designs and machine-readable interfaces.
- Use commodity hardware and certified COTS modules where sensible.
- Keep the safety case explicit, testable, and versioned with the design.
- Make local fabrication the default: cut, weld, bond, bolt, test.

Open

Readable source artifacts.

Repeatable

Regenerate cities, CAD, BOMs, docs.

Modular

Swap suppliers if envelopes fit.

Assessable

Evidence links to tests and drawings.

Design automation

GIS and city templates generate alignments, station placements, maps, cost summaries, and simulator scenarios.

Control software

Rust crates cover simulator, interlocking, ATP, brake, obstacle detection, TCMS, GUIs, and safety-case compilation.

Mechanical CAD

Parametric source generates trainsets, bogies, track, civil works, stations, depots, FreeCAD review files, and screenshots.

Hardware hosts

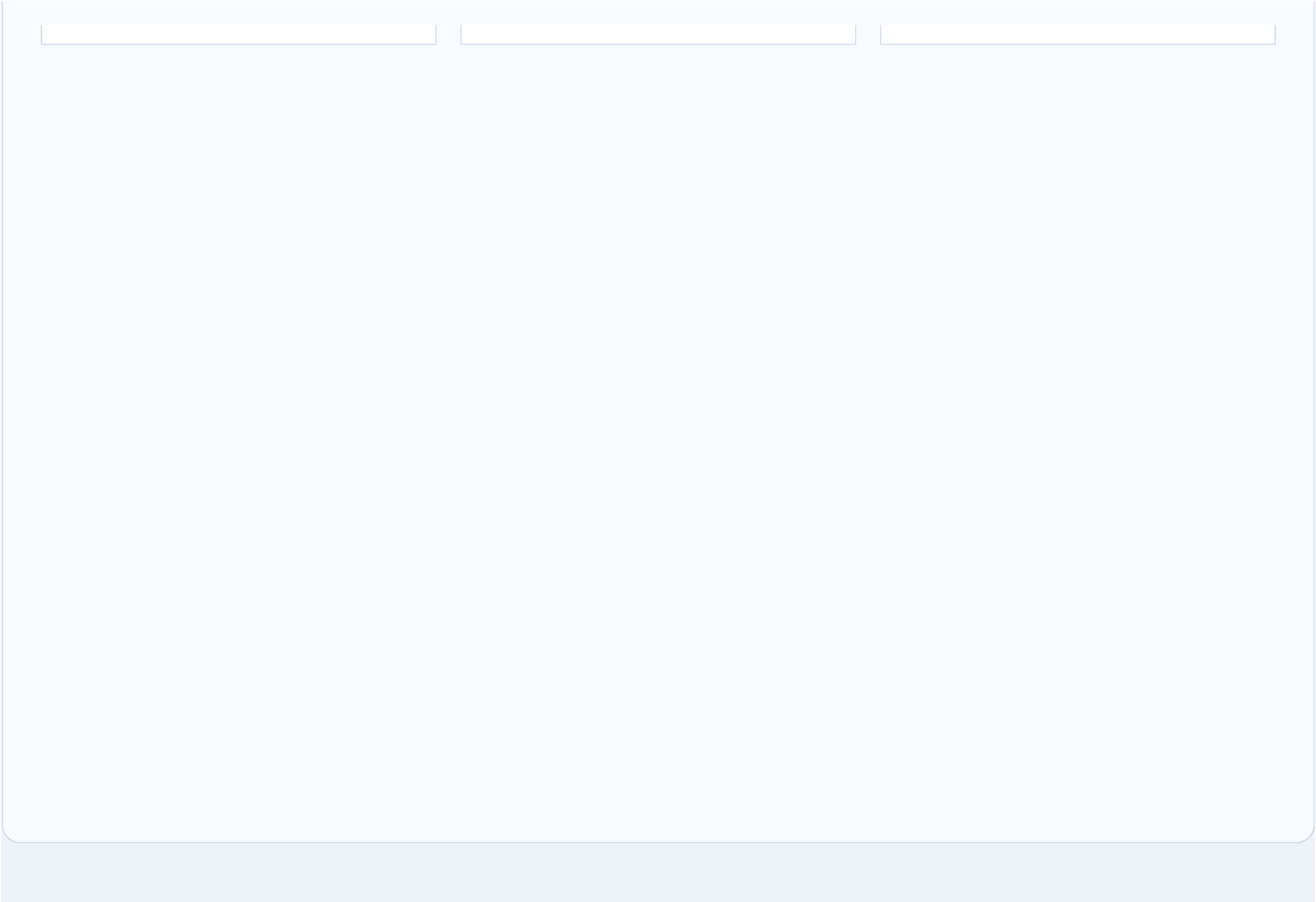
Reference SBC/ECU classes define train, wayside, station, and obstacle-detection hosts.

Operations docs

Rulebooks, staff procedures, maintenance flows, and depot processes live with the technical design.

Safety evidence

Certification and GSN documents capture assumptions, tests, and remaining evidence gaps.



2. Reference Vehicle

51 m

3-car reference consist over couplers.

17 m

Repeated car module for local fabrication.

12 doors

Two low-floor door pairs per side per car.

Passenger layout

- About 10 m low-floor centre section.
- PRM spaces and wheelchair turning circles in the centre zone.
- Longitudinal seats over under-seat sodium-ion batteries.

Running gear

- Standard bogies about 3 m long at car ends.
- High-floor end decks over bogies.
- Two powered bogies and four trailer bogies in the base consist.

Driverless ends

- Open glass passenger view, front and rear.
- LED headlamp and marker bars below the glass.
- T-OBS sensors behind serviceable cowl hardware.

Design authority

Parametric source in `mechanical-py/src/osr_mech/rolling_stock` is the basis. FreeCAD files and PNGs are generated review and handoff artifacts.

Procurement stance

Make the steel datum structure locally; use supplier-neutral envelopes for doors, windows, HVAC, bogies, traction, batteries, electronics, and passenger modules.

3. Trainset Layout



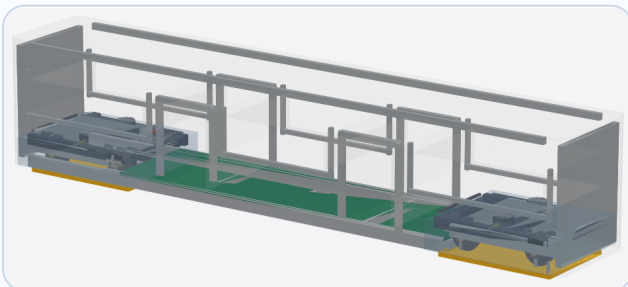
Why 17 m cars?

The 17 m module keeps a 3-car train at 51 m, leaves platform margin on a 67 m standard station, and remains workshop-friendly.

- Standard bogies sit at each car end.
- Raised decks cover bogie structure and service access.
- The low-floor middle supports doors, PRM bays, and passenger flow.
- Capacity scales by adding repeated cars or by a later 19-20 m variant.

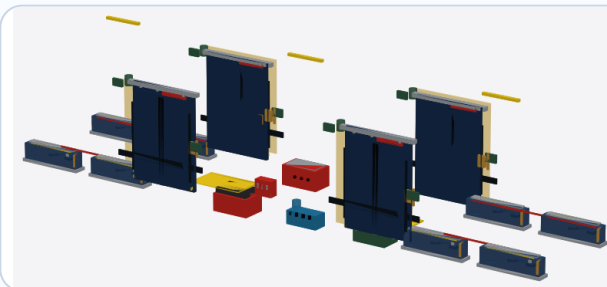
2 + 4

Two powered bogies, four trailer bogies.



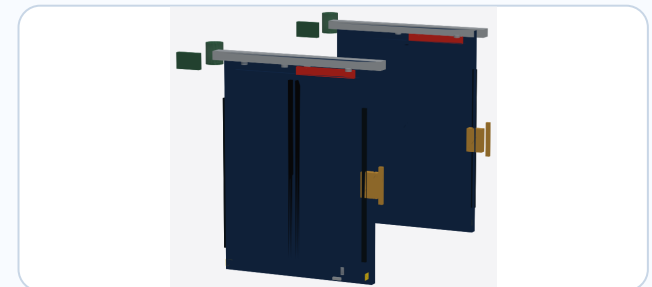
12

Exterior door cassettes per consist.



450 kWh

Reference sodium-ion onboard energy.



PRODUCTION ASSEMBLY

17 m modular solar metro car: fabricated frame + COTS modules

Assembly strategy

MAKE datum frame; use BID/COTS doors, glass, HVAC, bogies, electronics, and interiors.



Why this is easy to produce

Repeated 17 m module, few welded datum parts, late COTS installation, bolted service cassettes.
No bespoke extrusion dies; STEP output is generated from build123d source.

1 Weld Underframe In Fixture



S355 ladder frame, bolsters, coupler pockets, jacking pads. Flat-jig welding keeps tolerances.

2 Add Side/Roof Spaceframe



Door portals, window rails, centrails, end rings, and roof bows. Primary load path stays steel.

3 Install COTS Modules



Door/window cassettes, seats over batteries, roof PV/HVAC, wiring trays, and panels install late.

4 Lower Onto Bogies + Couple



Standard 3 m bogies, high-floor end decks, 10 m low-floor centre. Repeat cars make the trainset.

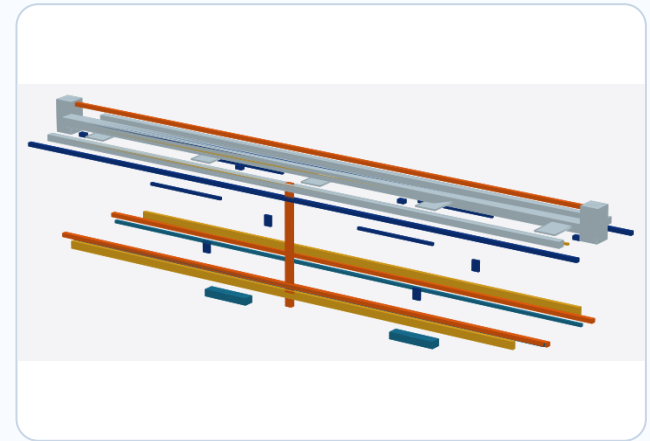
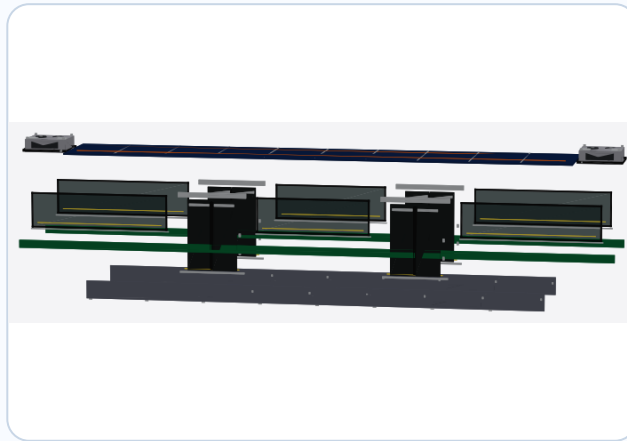
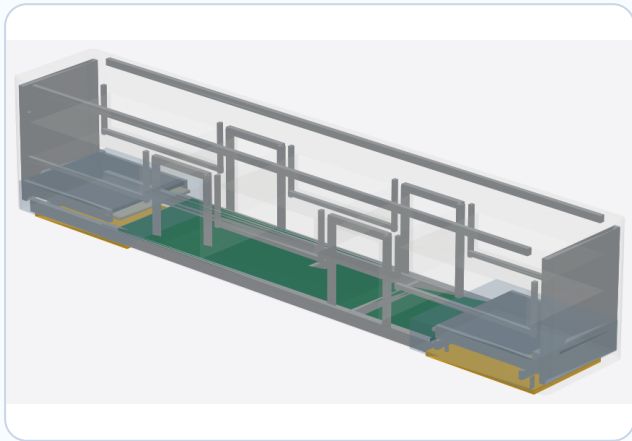


4. Layered Car Body

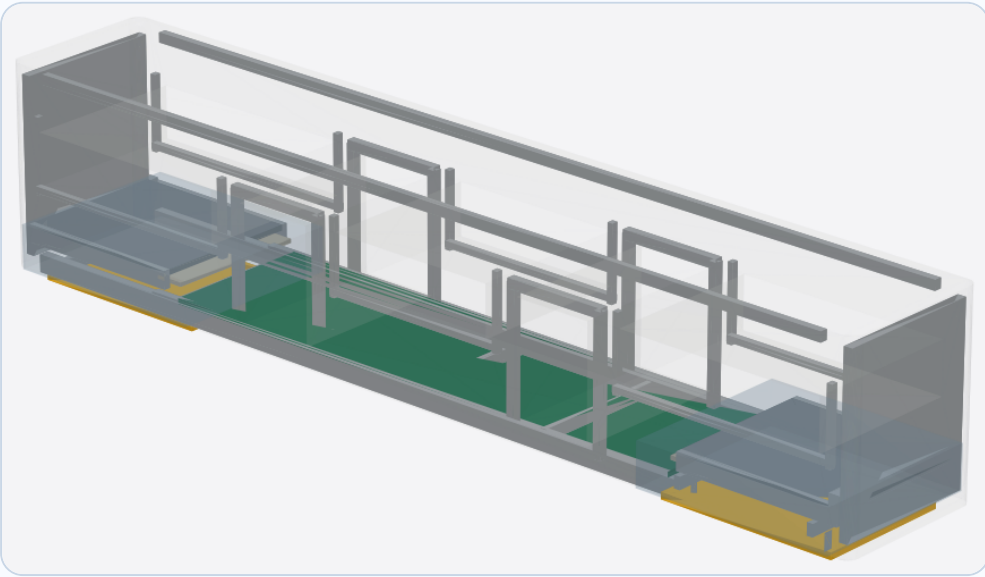


Components to final assembly

- Primary steel shell, floor, and portal structure.
- Exterior cladding, glazing, doors, livery, solar, and skirts.
- Interior fit-out with seats, PRM bays, grab rails, and displays.
- Service layers for HVAC, LV/data, HV traction, PV, coolant, and fire paths.



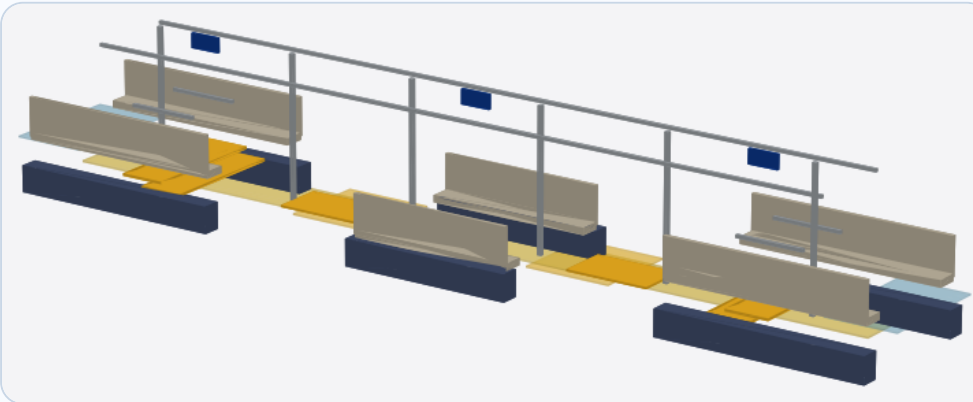
5. Body Structure And Load Path



Fabricated, inspectable steel

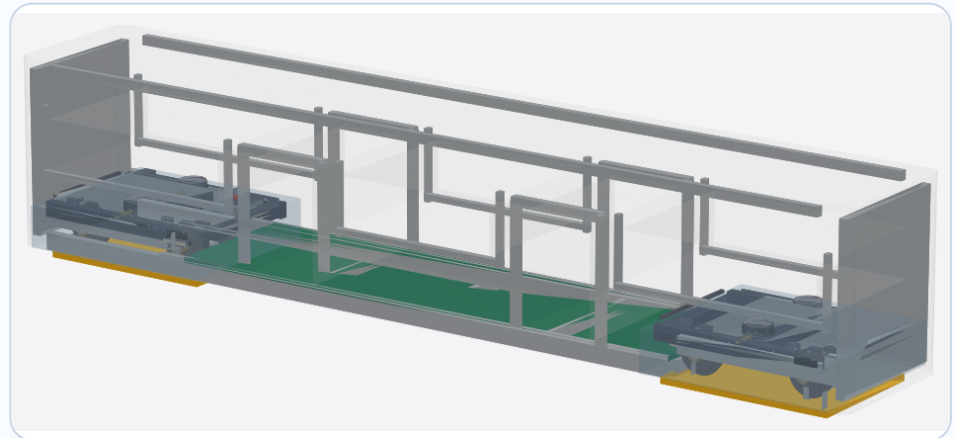
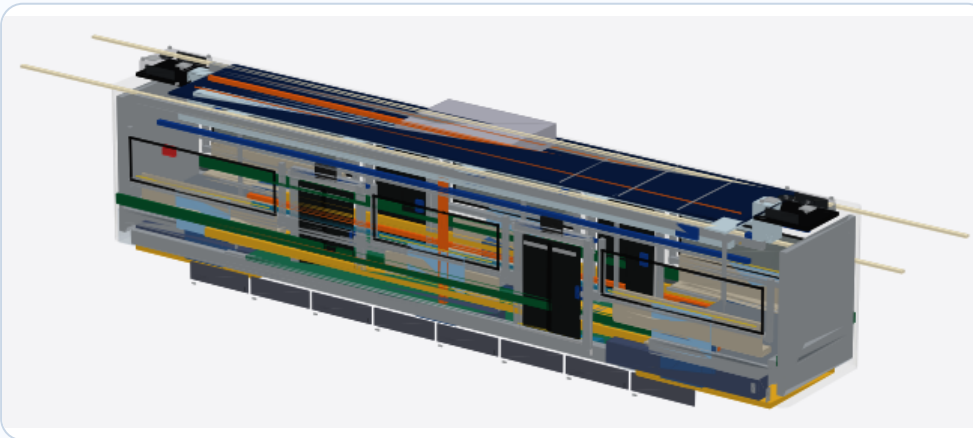
- S355 underframe ladder and side/roof spaceframe.
- Door portal posts and headers around COTS cassettes.
- Window posts, waist rails, cantrails, and end rings.
- Anti-climber beam and open-glass cowl structural ring.
- Bogie rotation, suspension-travel, and drop-zone envelopes.

6. Access And Interior



Passenger design intent

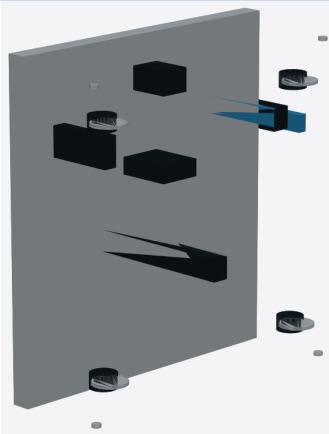
- Two door pairs per side per car, concentrated in the low-floor zone.
- Wheelchair turning circles and PRM bays in the centre section.
- Longitudinal seats over under-seat battery strakes.
- Open passenger view through both driverless glass ends.
- Handrails and step treads where the floor rises over bogies.



7. Driverless Ends And Sensing

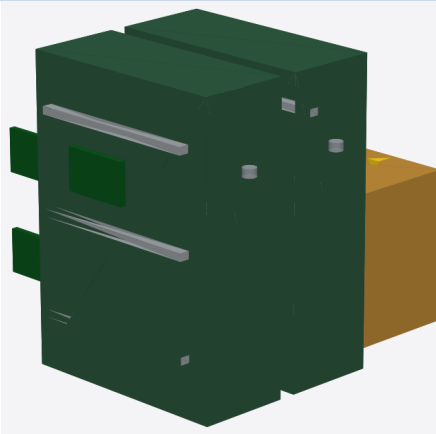
Open glass ends

The ends are not cab partitions. Passengers retain a forward/rear view through heated RF-transparent glass.



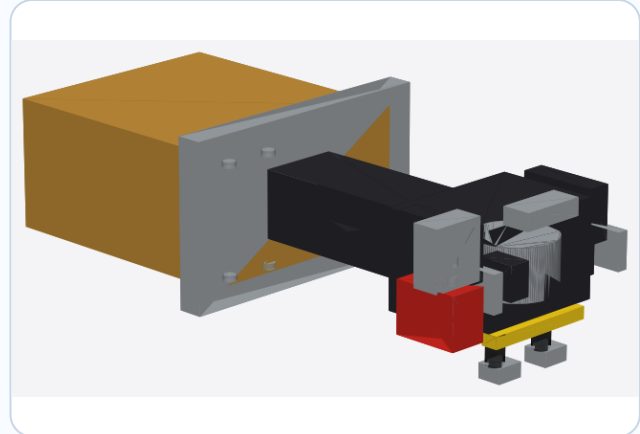
T-OBS sensor pack

LIDAR, mmWave radar, stereo cameras, and ultrasonic sensors sit behind serviceable cowl hardware.

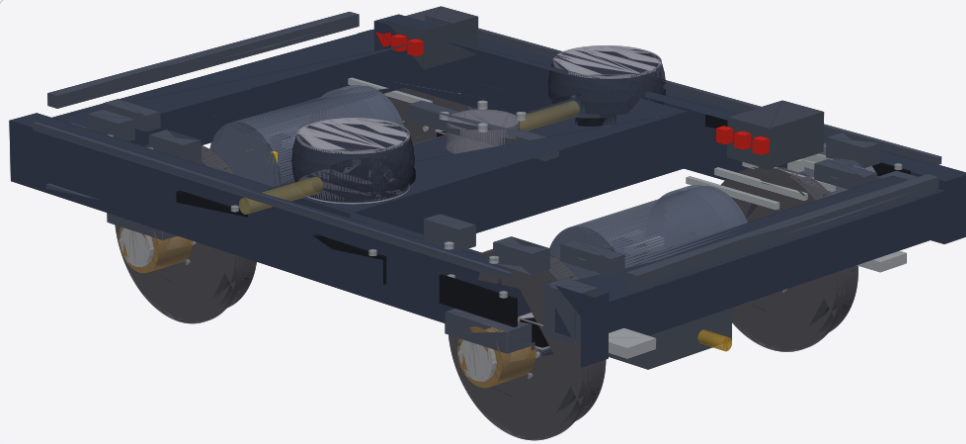


Recovery only

Hidden maintenance/recovery controls are not a normal driver cab.

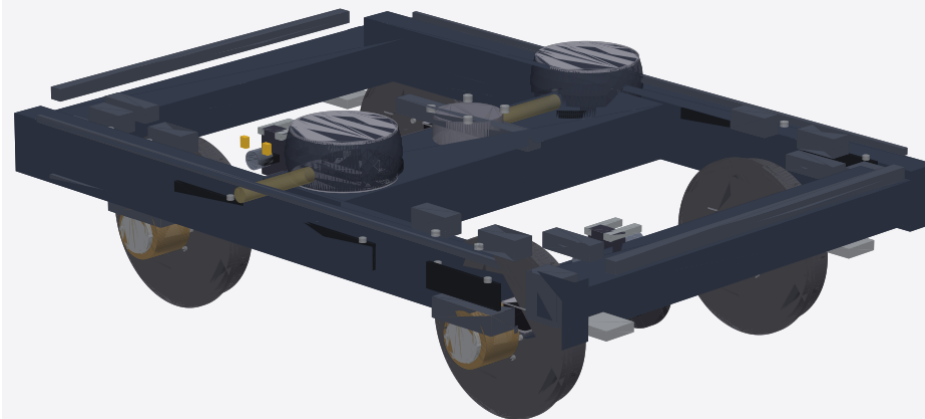


8. Bogies And Running Gear

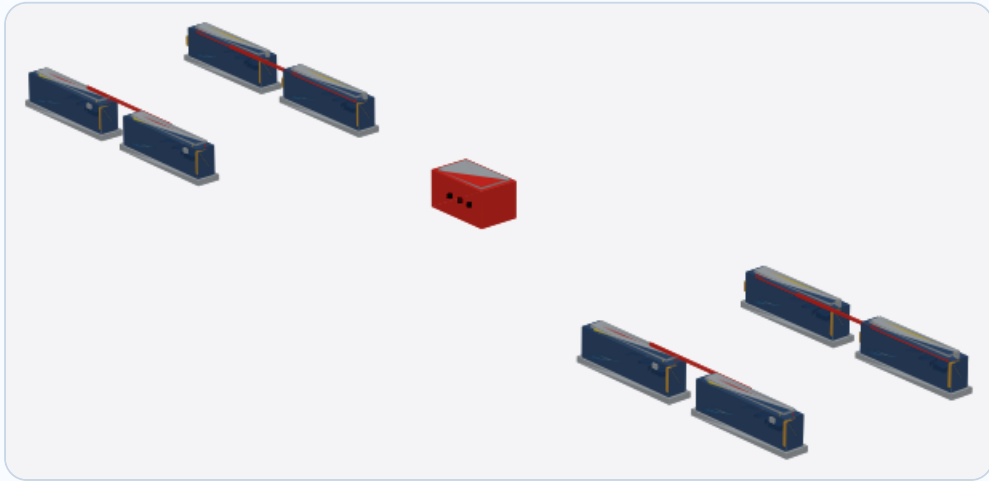


Standardized bogie strategy

- Two powered bogies total, at the outer end cars.
- Four trailer bogies using the common frame and suspension envelope.
- Powered bogie includes PMSM motors, gearboxes, brakes, and suspension.
- Trailer bogie keeps the same maintainable running-gear architecture without traction drives.

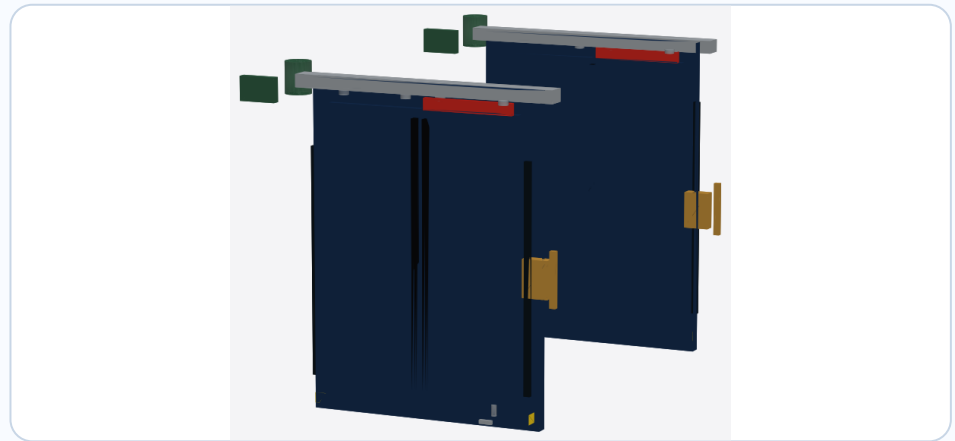
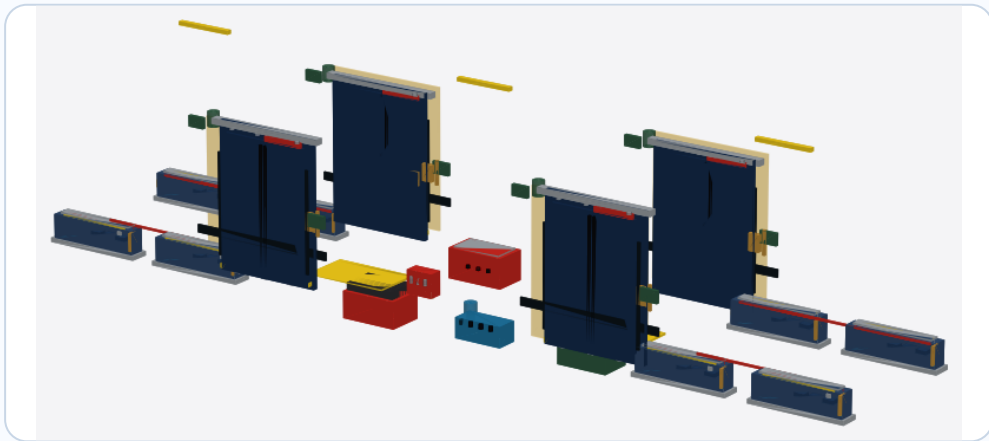


9. Energy System

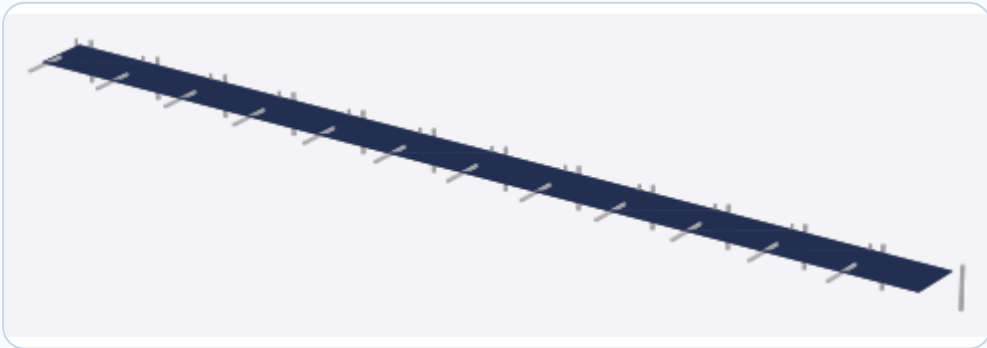


Battery electric, catenary-free

- Reference 450 kWh sodium-ion consist energy.
- Under-seat modules keep the roof available for solar and HVAC.
- Station side-pin charging is the primary concept.
- Roof PV supports auxiliary energy and site energy modelling.



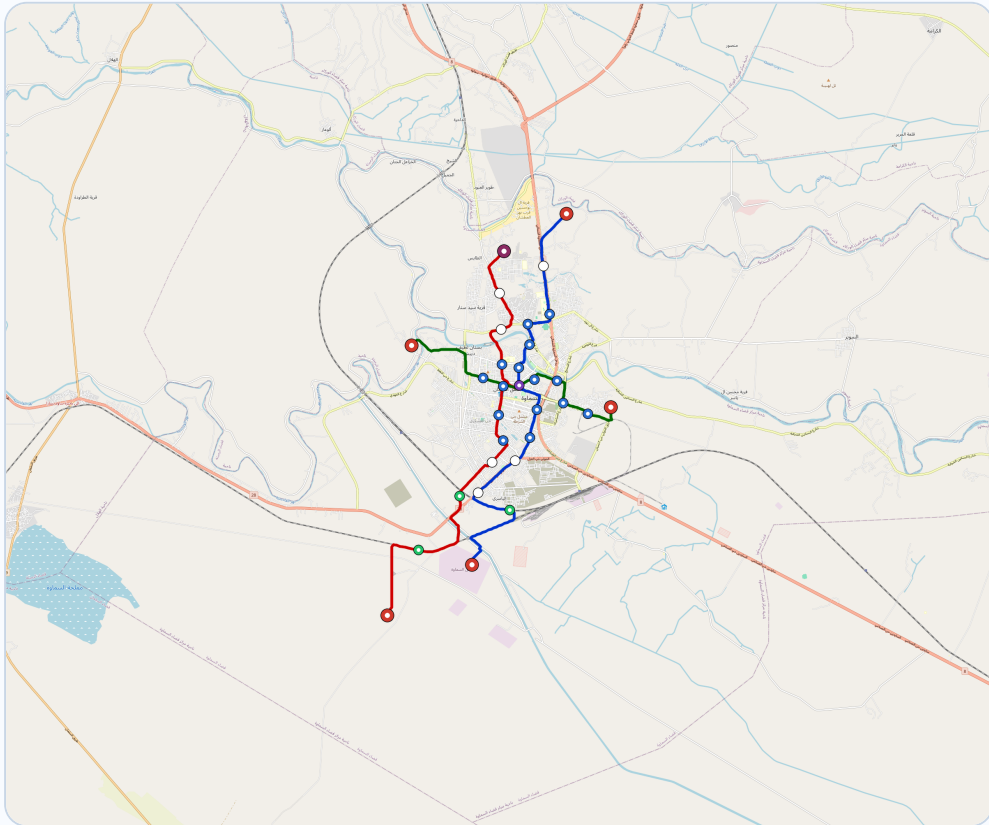
10. Stations And Civil Works



Station pattern

- Low platform interface at the 350 mm train floor datum.
- Platform screen-door alignment and train-side interlocks are reserved in CAD.
- Station charging can be side-pin or pantograph-down depending on civil geometry.
- Stations, track, and city designs regenerate from machine-readable templates.

11. City Design Automation



From city inputs to runnable scenarios

- World sample batches define candidate cities.
- GIS tooling emits route GeoJSON, maps, quality YAML, and README summaries.
- Rust design synthesis produces simulator scenarios and cost summaries.
- Every city folder is a reproducible design package.

12. Simulator And Control Stack



Software crates

- Simulator and scenario execution.
- Interlocking, ATP, brake, and obstacle-detection crates.
- TCMS and GUI tooling.
- Safety-case compiler and verification support.



13. Hardware Host Classes

T-ECU/S

Train safety kernel host for safety-related train functions.

T-ECU/A

Train application computer for non-safety application logic and integrations.

T-OBS

Obstacle-detection host connected to the driverless nose sensor pack.

W-SBC

Wayside controller class for station and trackside equipment.

S-SBC

Station/depot host for local operations and maintenance integration.

DIY pilots

Commodity-module assembly path for lab and early integration builds.

14. Safety And Certification

Safety case is part of the product

The repository includes RFCs, safety-case documents, certification packs, operations rules, and formal models. The project does not certify deployments; it produces artifacts suitable for independent assessment.

Evidence direction

- Requirements and assumptions are explicit.
- Tests and generated artifacts are linked to source.
- Supplier documents fit controlled interface envelopes.
- Authority review remains deployment-specific.

Concept

Reference architecture and RFCs.

v1

Envelope CAD, BOM, docs, simulation.

v2

Supplier envelopes and shop drawings.

First article

Build, inspect, static test.

Assessment

Independent review and authority process.

15. Manufacturing, Cost, Operations

Manufacturing doctrine

Scope	Default	Reason
Primary structure	MAKE: cut, bend, weld S355 steel	Common regional workshop capability.
Panels	BID/SOURCE composite panels	Low tooling cost and replaceable after damage.
Doors, windows, HVAC	COTS or BID modules	Use certified rail/bus evidence instead of bespoke development.
Bogies	Qualified MAKE plus certified rotating/brake parts	Local frame production with supplier evidence for safety-critical parts.
Traction and BMS	BID/SOURCE packages	Supplier qualification, EMC, and lifecycle test evidence required.

Planning-grade cost direction

\$2.21M

Direct material planning estimate for 3-car consist.

\$2.98M

Planning-grade consist cost after shop labor uplift.

12

COTS electric door cassettes per consist.

18

COTS window cassettes per consist.

MAKE

Fabricator-owned steel datum parts, brackets, harness supports, and local tooling.

BID/SOURCE

Doors, windows, HVAC, electronics, traction, bogie parts, and passenger modules.

Operations

Rulebook, OCC guidance, station staff procedures, and depot flows are kept with the technical design.

Maintenance

Bolted service cassettes, removable skirts, access lids, and supplier module replacement.

Recovery

Rescue couplers, recovery cabinets, and explicit degraded-mode procedures for driverless service.

16. Deployment Workflow And Repository Map

1. Select corridor

Use city generation and GIS evidence.

2. Configure system

Pick consist, station, depot, energy, and operations templates.

3. Freeze envelopes

Supplier modules must fit reserved interfaces.

4. Build first article

Fabricate, install COTS modules, inspect.

5. Certify locally

Submit evidence to authority and independent assessor.

Repository map

Path	Purpose
crates/	Rust simulator, control, safety, GUI, and design tools.
design-py/ and designs/	GIS sidecar and generated city design packages.
mechanical-py/	Parametric mechanical source and generated FreeCAD review artifacts.
hardware/	Hardware references and host-class integration.
docs/	Architecture, RFCs, operations, certification, safety case, rolling-stock docs.
formal/	TLA+ consensus specification and model-checking harnesses.

Regenerable artifacts

City packages

Route GeoJSON, maps, quality YAML, simulator scenarios, and README summaries.

Mechanical outputs

CAD screenshots, FreeCAD review files, rolling-stock BOM, and manufacturing views.

Software evidence

Tests, simulator runs, formal models, and safety-case traces.

Reader editions

Brochures, documentation books, and repository indices.

17. Regeneration Commands And Evidence Hooks

Repository health

```
python3 scripts/repo-health.py --quiet
```

Mechanical tests

```
PYTHONPATH=mechanical-py/src pytest  
mechanical-py/tests -q
```

CAD screenshots

```
PYTHONPATH=mechanical-py/src  
python3 mechanical-py/scripts/  
render_screenshots.py
```

FreeCAD review files

```
PYTHONPATH=mechanical-py/src  
mechanical-py/scripts/  
freecad_trainset.sh --family light-  
metro-3car
```

BOM CSV

```
python3 scripts/export-light-metro-  
bom.py
```

City regeneration

```
scripts/regenerate-city.sh samawah
```

Tests

Unit, integration, simulator, and formal checks identify artifact drift.

Drawings

v2 drawing register turns envelope CAD into shop-release control.

Evidence

Certification and safety-case folders record assumptions and gaps.

Evidence release chain

Design freeze

Generated assemblies, BOM, screenshots, and supplier envelope checks.

Simulation freeze

Routes, stations, headways, degraded modes, and reproduced scenario logs.

Hardware freeze

Host class, IO map, watchdog policy, and safety boundary assumptions.

Assessment pack

Hazards, mitigations, open evidence requests, and authority handoff notes.

Release note

What changed, which artifacts regenerated, and which checks passed.

18. v0.2 Focus And Next Reading

Rolling stock detail design

- Supplier exact envelopes for doors, windows, HVAC, traction, and bogie components.
- 2D shop drawings, weld maps, tolerance stacks, and inspection plans.
- Harness clamp locations, maintainability clearances, FEA-ready brackets, and drawing control.

System integration

- Station charging interface and platform screen-door proof.
- Hardware bring-up and simulated end-to-end control paths.
- Expanded certification evidence, test traceability, generated cities, and cost comparisons.

Core entry points

- README.md for the repository overview.
- docs/ARCHITECTURE.md for system structure.
- docs/rolling-stock/light-metro-3car/README.md for the reference train.
- docs/rfcs/README.md for design decisions.
- docs/certification/ and docs/safety-case/ for assessment evidence.

Open artifact, not a certifier

OpenSourceRail produces open designs, software, hardware references, and evidence packs. Deployment partners and national authorities remain responsible for supplier qualification, safety assessment, homologation, and operation.

Start with the generated trainset and production graphics, then follow the parametric source.

v0.2 deliverable checklist

Rolling stock CAD

Exact supplier envelopes and 2D shop drawing register.

Body and chassis

Bend sequence, weld map, access panels, and datum scheme.

Electrical and HVAC

Harness zones, duct routes, service clearances, and module boundaries.

Deployment evidence

City comparisons, cost sensitivity, and assumption ledger.